NEET-UG CHEMISTRY SPECIALIZATION

Unit: Biomolecules

Complete NCERT-Aligned Theory, Chemical Transformations, Structural Visualizations, and High-Yield Assessment

1. NEET Exam Context & Learning Objectives

In living organisms, non-living chemical components assemble dynamically into structures capable of growth, sustainment, and genetic reproduction. This field of chemical inquiry represents **Biochemistry**. Biomolecules—including carbohydrates, proteins, enzymes, vitamins, nucleic acids, and hormones—constitute the fundamental molecular logic of physiological processes.

Core NEET-UG Learning Objectives:

- Delineate the structural classification and chemical evidence establishing the open and cyclic hemiacetal pyranose forms of D-(+)-glucose and the furanose forms of D-(-)-fructose.
- Identify glycosidic bonds in disaccharides (sucrose, maltose, lactose) and polysaccharides (amylose, amylopectin, cellulose, glycogen), predicting reducing and non-reducing characteristics.
- Characterize α-amino acids, their zwitterionic dipolar equilibria, isoelectric properties, and essential versus non-essential categorizations.
- Distinguish the four structural levels of proteins ($1^\circ, 2^\circ, 3^\circ, 4^\circ$), differentiate fibrous from globular architectures, and interpret molecular denaturation mechanisms.
- Quantify biocatalysis by enzymes via activation energy reduction.
- Classify vitamins by solubility, storage, dietary sources, and deficiency pathologies.
- Master nucleotide composition (purines, pyrimidines, pentose sugars, phosphate), $3' \rightarrow 5'$ phosphodiester bridges, Watson-Crick double-helical complementary base pairing, RNA forms, and forensic DNA fingerprinting.
- Categorize endocrine hormones by chemical origin (steroids, polypeptides, amino acid derivatives) and understand their homeostatic regulatory roles.

2. Carbohydrates

Carbohydrates are naturally occurring organic compounds predominantly synthesized by plants via photosynthesis. Historically, they were regarded as hydrates of carbon obeying the empirical formula $C_x(H_2O)_y$. For instance, glucose ($C_6H_{12}O_6$) matches $C_6(H_2O)_6$. However, this classical definition fails:

- Acetic acid ($CH_3COOH \equiv C_2H_4O_2 \equiv C_2(H_2O)_2$) fits the formula but exhibits pure carboxylic acid characteristics, not carbohydrate behavior.
- Rhamnose ($C_6H_{12}O_5$) and 2-deoxyribose ($C_5H_{10}O_4$) are verified carbohydrates that violate the empirical formula.

Modern Chemical Definition: Optically active polyhydroxy aldehydes or ketones, or substances that produce such units upon aqueous hydrolysis.

2.1 Classification Based on Hydrolytic Behavior

1. **Monosaccharides:** Simplest monomeric carbohydrates that cannot be hydrolysed further into smaller polyhydroxy aldehyde or ketone units. About 20 exist in nature (e.g., glucose, fructose, ribose, galactose).
2. **Oligosaccharides:** Carbohydrates that yield 2 to 10 monosaccharide units upon hydrolysis. Sub-classified by monomer count into disaccharides (e.g., sucrose, maltose, lactose), trisaccharides (e.g., raffinose), etc. Disaccharides are the most prevalent.
3. **Polysaccharides:** High-molecular-mass polymers that yield a large number of monosaccharide units on complete hydrolysis (e.g., starch, cellulose, glycogen). Because they lack sweet taste, they are termed *non-sugars*.

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2.2 Reducing vs. Non-Reducing Sugars

Sugars that reduce Fehling's solution (giving a red precipitate of Cu_2O) and Tollens' reagent (forming a silver mirror) are designated **reducing sugars**.

- **All monosaccharides**—whether aldoses or ketoses—are reducing sugars. (Ketoses reduce Tollens' and Fehling's reagents because they undergo α-hydroxyketone tautomeric rearrangement to aldoses via an enediol intermediate in alkaline test solutions).
- In disaccharides, if the anomeric reducing centers (hemiacetal/hemiketal carbons) are engaged in glycosidic linkage, the sugar is **non-reducing** (e.g., sucrose). If at least one anomeric carbon retains a free hemiacetal group capable of mutarotating to an open aldehyde, it is a **reducing sugar** (e.g., maltose, lactose).

Monosaccharide Classification Based on Carbon Count and Carbonyl Group

Carbon Count	General Class	Aldehyde Variant (Aldose)	Ketone Variant (Ketose)
3	Triose	Aldotriose (Glyceraldehyde)	Ketotriose (Dihydroxyacetone)
4	Tetrose	Aldotetrose (Erythrose)	Ketotetrose (Erythrulose)
5	Pentose	Aldopentose (Ribose, Xylose)	Ketopentose (Ribulose)
6	Hexose	Aldohexose (Glucose, Galactose)	Ketohexose (Fructose)
7	Heptose	Aldoheptose	Ketoheptose

2.3 D-(+)-Glucose: Preparation and Proof of Constitution

Glucose (aldohexose, dextrose) is the monomer of starch and cellulose. It is prepared via two standard pathways:

1. Laboratory Hydrolysis of Sucrose (Cane Sugar):

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \xrightarrow[\text{alcoholic medium}]{\mathrm{H^+}} C_6H_{12}O_6\ (\text{D-glucose}) + C_6H_{12}O_6\ (\text{D-fructose})}$$

2. Industrial Hydrolysis of Starch:

$$\mathrm{(C_6H_{10}O_5)_n + nH_2O \xrightarrow[393\ K,\ 2\text{--}3\ atm]{\text{dil. } H_2SO_4} nC_6H_{12}O_6\ (\text{D-glucose})}$$

Chemical Reactions Establishing Glucose Structure

1. **Molecular Formula:** Elemental analysis and cryoscopic molecular mass determination reveal $\mathrm{C_6H_{12}O_6}$.
2. **Straight 6-Carbon Chain:** Prolonged heating of glucose with hydrogen iodide (HI) and red phosphorus at elevated temperature yields exclusively n-hexane:
   $$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH \xrightarrow{HI,\ \Delta} CH_3\text{--}CH_2\text{--}CH_2\text{--}CH_2\text{--}CH_2\text{--}CH_3}$$
   This confirms all six carbons are connected in a continuous, unbranched sequence.
3. **Carbonyl Group Confirmation:**
   - Reacts with hydroxylamine ($\mathrm{NH_2OH}$) to form a stable *oxime*:
     $$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH + NH_2OH \rightarrow CH=N\text{--}OH\text{--}(CHOH)_4\text{--}CH_2OH + H_2O}$$
   - Adds hydrogen cyanide (HCN) to yield a *cyanohydrin*:
     $$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH + HCN \rightarrow CH(OH)(CN)\text{--}(CHOH)_4\text{--}CH_2OH}$$

   These nucleophilic additions corroborate the presence of a carbonyl ($>\mathrm{C=O}$) function.
4. **Aldehyde Group Nature:** Glucose reacts with a mild oxidising agent like bromine water ($\mathrm{Br_2/H_2O}$) to form a six-carbon monocarboxylic acid, **gluconic acid**:
   $$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH \xrightarrow{Br_2\ water} COOH\text{--}(CHOH)_4\text{--}CH_2OH}$$
   Because bromine water oxidises aldehydes without attacking ketones or secondary alcohols, the carbonyl must be terminal ($-\mathrm{CHO}$).
5. **Presence of Five Hydroxyl Groups:** Acetylation with acetic anhydride ($\mathrm{Ac_2O}$) in the presence of pyridine yields **glucose pentaacetate**:
   $$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH + 5(CH_3CO)_2O \rightarrow CHO\text{--}[CH(OCOCH_3)]_4\text{--}CH_2OCOCH_3 + 5CH_3COOH}$$
   Because geminal diols are unstable and dehydrate spontaneously to carbonyls, each of the five $-\mathrm{OH}$ groups must reside on a different carbon atom.
6. **Presence of a Primary Alcoholic Group:** Vigorous oxidation of glucose or gluconic acid with concentrated nitric acid ($\mathrm{HNO_3}$) yields the six-carbon dicarboxylic acid **saccharic acid** (glucaric acid):

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$$\mathrm{CHO\text{--}(CHOH)_4\text{--}CH_2OH} \xrightarrow{\mathrm{HNO_3,\ [O]}} \mathrm{COOH\text{--}(CHOH)_4\text{--}COOH} \xleftarrow{\mathrm{HNO_3,\ [O]}} \mathrm{COOH\text{--}(CHOH)_4\text{--}CH_2OH}$$

Oxidation of both the terminal aldehyde and the opposite terminus into carboxyl groups proves that the sixth carbon atom bears a primary alcohol group ($-\mathrm{CH_2OH}$).

2.4 Fischer Projection and Relative D/L Configuration

Carbohydrate stereochemistry is standardized against the reference standard **glyceraldehyde**. In Fischer projections oriented with the most oxidized carbon ($-\mathrm{CHO}$) at the top:

- If the hydroxyl group at the lowest asymmetric center (carbon-5 in aldohexoses) projects to the **right**, the molecule possesses the D**-configuration**.
- If this hydroxyl group projects to the **left**, it has the L**-configuration**.
- The symbols D and L indicate relative spatial stereochemical configuration and must not be confused with optical rotation symbols $(+)$ [dextrorotatory] or $(-)$ [laevorotatory], which describe physical polarimetric rotation. Glucose is naturally D-(+)-glucose.

2.5 Cyclic Structure of Glucose and Anomerism

The open-chain Fischer aldehyde structure fails to explain three fundamental experimental observations:

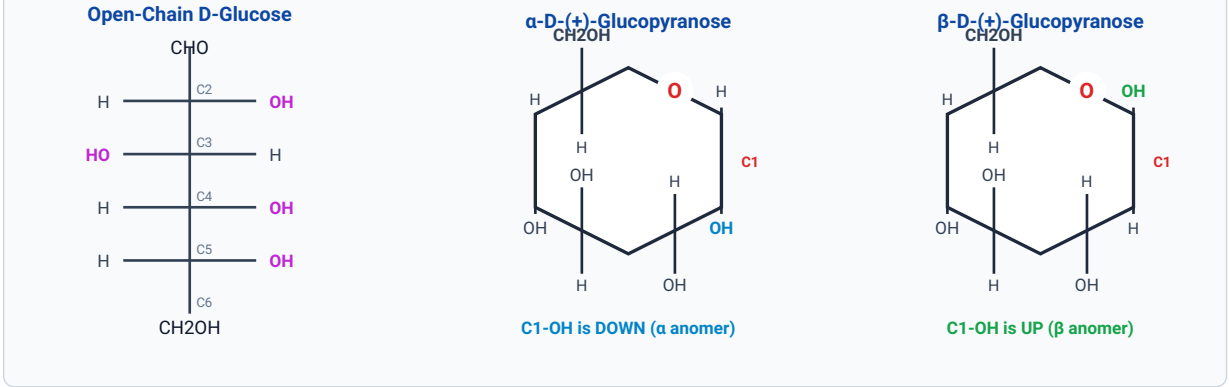
Limitations of the Open-Chain Aldehyde Model:

1. Despite bearing an aldehyde function, glucose does not restore the magenta color of Schiff's reagent and fails to form the sodium bisulphite ($\mathrm{NaHSO_3}$) adduct.
2. Glucose pentaacetate fails to react with hydroxylamine ($\mathrm{NH_2OH}$), indicating the complete absence of an accessible free $-\mathrm{CHO}$ group in the acetylated form.
3. Glucose exists as two distinct, isolable crystalline diastereomers designated α and β:
   - α**-form:** Melting point $419\ \mathrm{K}$ (specific rotation $+112^\circ$), obtained by crystallization from concentrated solution at $303\ \mathrm{K}$.
   - β**-form:** Melting point $423\ \mathrm{K}$ (specific rotation $+19^\circ$), obtained by crystallization from hot, saturated aqueous solution at $371\ \mathrm{K}$.

These observations proved that glucose forms an intramolecular cyclic **hemiacetal**. The nucleophilic $-\mathrm{OH}$ group on carbon-5 adds across the carbon-1 carbonyl group, generating a stable, unstrained six-membered heterocyclic ring (termed **pyranose**, by analogy with pyran).

During cyclization, carbon-1—originally achiral in the open chain—becomes asymmetric. It is termed the **anomeric carbon**. The two resulting stereoisomers, differing solely in the configuration at $\mathrm{C1}$, are called **anomers**:

- α**-D-(+)-glucopyranose:** The anomeric $\mathrm{C1}$ hydroxyl group lies on the *right* in the Fischer ring (pointing *downward* below the ring plane in Haworth projection).
- β**-D-(+)-glucopyranose:** The anomeric $\mathrm{C1}$ hydroxyl group lies on the *left* in the Fischer ring (pointing *upward* above the ring plane in Haworth projection).



Structural Formulations of D-Glucose: Fischer Open Chain vs. Haworth Pyranose Projections of α and β Anomers

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2.6 D-(-)-Fructose

Fructose is a natural ketohexose occurring in honey, sweet fruits, and vegetables. With molecular formula $\mathrm{C_6H_{12}O_6}$, it possesses a ketonic group at carbon-2 and an unbranched six-carbon chain. Fructose is laevorotatory, designated **D-(-)-fructose**.

In solution, the $-\mathrm{OH}$ at carbon-5 adds intramolecularly across the $\mathrm{C2}$ ketone, forming a five-membered cyclic hemiketal called **furanose** (by analogy with furan). Its anomers are:

- **α-D-(-)-fructofuranose:** Anomeric $\mathrm{C2}$ bears $-\mathrm{OH}$ pointing downward in Haworth projection.
- **β-D-(-)-fructofuranose:** Anomeric $\mathrm{C2}$ bears $-\mathrm{OH}$ pointing upward in Haworth projection.

2.7 Disaccharides and the Glycosidic Linkage

Disaccharides are formed when two identical or distinct monosaccharide units condense with the loss of one water molecule. The covalent oxide bridge connecting the two monosaccharides through an oxygen atom is termed a **glycosidic linkage**.

1. Sucrose ($\mathrm{C_{12}H_{22}O_{11}}$)

- Composed of α-D-glucopyranose and β-D-fructofuranose.
- **Linkage:** Glycosidic linkage connects $\mathrm{C1}$ of α-D-glucose to $\mathrm{C2}$ of β-D-fructose.
- **Non-reducing Sugar:** Because both anomeric reducing centers ($\mathrm{C1}$ of glucose and $\mathrm{C2}$ of fructose) are bonded in the glycosidic bridge, sucrose cannot open into a free carbonyl form. It does not reduce Tollens' or Fehling's reagents and does not undergo mutarotation.
- **Invert Sugar:** Native sucrose is dextrorotatory, with a specific rotation of $+66.5^\circ$. Upon acid-catalysed or enzymatic (invertase) hydrolysis, it produces an equimolar mixture of D-(+)-glucose ($[\alpha]_\mathrm{D} = +52.5^\circ$) and D-(-)-fructose ($[\alpha]_\mathrm{D} = -92.4^\circ$):

$$\text{Net Rotation of Hydrolysate} = \frac{+52.5^\circ + (-92.4^\circ)}{2} = -19.95^\circ$$

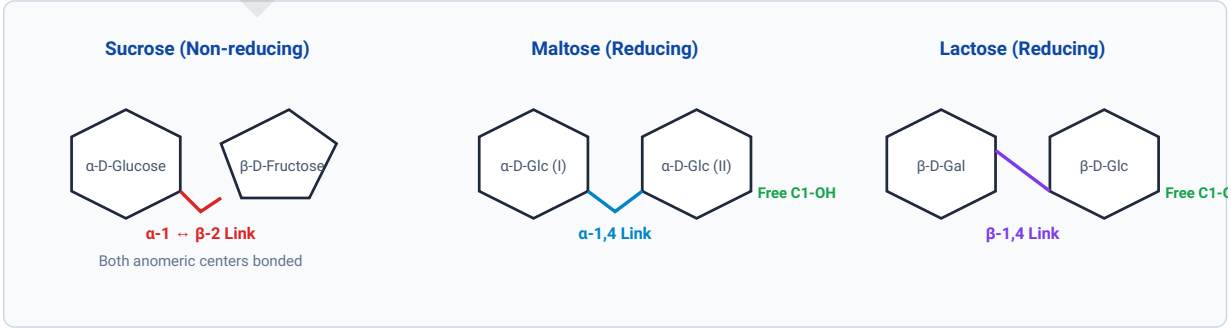
Because the intense laevorotation of fructose exceeds the dextrorotation of glucose, the optical rotation inverts from positive ($+$) to negative ($-$). The resulting hydrolysed mixture is known as **invert sugar**.

2. Maltose ($\mathrm{C_{12}H_{22}O_{11}}$)

- Composed of two α-D-glucopyranose units.
- **Linkage:** α-1,4-glycosidic bond joining $\mathrm{C1}$ of the first glucose unit to $\mathrm{C4}$ of the second.
- **Reducing Sugar:** The second glucose unit retains an intact hemiacetal group at its anomeric $\mathrm{C1}$, which can open in aqueous solution to yield a free aldehyde. Hence, maltose reduces Fehling's and Tollens' reagents.

3. Lactose ($\mathrm{C_{12}H_{22}O_{11}}$) — Milk Sugar

- Composed of β-D-galactopyranose and β-D-glucopyranose.
- **Linkage:** β-1,4-glycosidic linkage connecting $\mathrm{C1}$ of β-D-galactose to $\mathrm{C4}$ of β-D-glucose.
- **Reducing Sugar:** The $\mathrm{C1}$ carbon of the glucose ring retains a hemiacetal group capable of generating a free aldehyde in solution.



Comparative Glycosidic Connectivity: Non-reducing Sucrose (α-$1 \leftrightarrow \beta$-2) vs. Reducing Maltose (α-$1,4$) and Lactose (β-$1,4$)

2.8 Polysaccharides: Starch, Cellulose, and Glycogen

Comparative High-Yield Structural Matrix of Natural Polysaccharides

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Polymer	Monomer Unit	Chain Architecture	Glycosidic Linkages	Water Solubility & Functions
Amylose (15–20% of starch)	α-D-(+)-Glucose	Unbranched, linear helix ($200\text{--}1000$ units)	α-1,4-glycosidic only	Soluble in water; compact dietary plant energy storage.
Amylopectin (80–85% of starch)	α-D-(+)-Glucose	Highly branched macromolecule	α-1,4 main chains; α-1,6 branches every $24\text{--}30$ units	Insoluble in water; primary dietary carbohydrate storage in plants.
Cellulose	β-D-(+)-Glucose	Completely straight, unbranched linear chains packing into rigid fibrils	β-1,4-glycosidic only	Insoluble; chief structural material of plant cell walls and cotton fibres.
Glycogen ("Animal Starch")	α-D-(+)-Glucose	Extremely branched (more dense than amylopectin)	α-1,4 chains with α-1,6 branch points every $8\text{--}12$ units	Soluble storage polymer concentrated in animal liver, muscle, and brain.

2.9 Biological Importance of Carbohydrates

- Primary metabolic fuel yielding ATP via cellular respiration in mammals.
- Major storage reserves: starch in plants and glycogen in animals. When physiological glucose demand rises, enzymes hydrolyse glycogen back to free glucose.
- Structural constituents: cellulose forms rigid bacterial and plant cell walls, timber, and cotton fibre.
- Genetic architecture: the pentoses D-ribose and 2-deoxy-D-ribose form the invariant structural backbone of RNA and DNA.

3. Amino Acids, Peptides, and Proteins

Proteins (Greek *proteios*, meaning primary or of prime importance) are the most abundant biomolecules in the living cell. They occur in every part of the organism, executing structural, transport, catalytic, and immunological functions. Proteins are linear heteropolymers of **α-amino acids**.

3.1 Structure, Stereochemistry, and Zwitterion Character

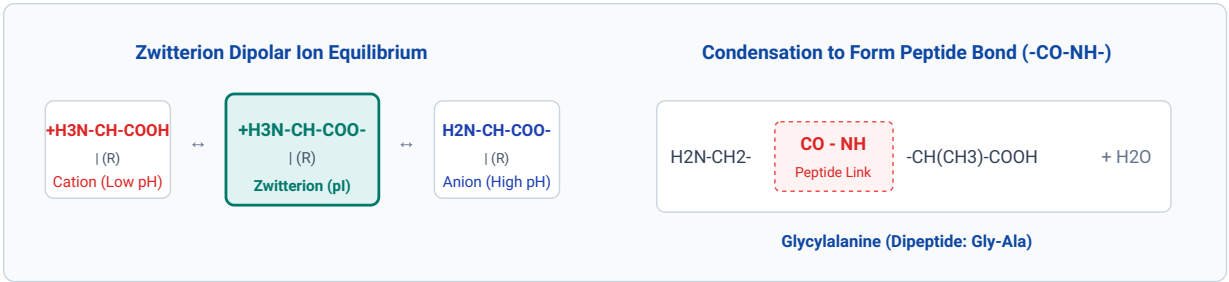
An α-amino acid possesses an amino group ($-\mathrm{NH_2}$) and a carboxyl group ($-\mathrm{COOH}$) bonded to the same α-carbon atom: $\mathrm{R\text{--}CH(NH_2)\text{--}COOH}$.

- **Chirality:** With the single exception of **glycine** ($\mathrm{R = H}$), the α-carbon of all naturally occurring amino acids is asymmetric (chiral). Hence, 19 of the 20 standard amino acids are optically active.
- **Configuration:** Naturally occurring amino acids predominantly exhibit the **L-configuration**, which means the $-\mathrm{NH_2}$ group projects to the left when represented in Fischer projection with $-\mathrm{COOH}$ oriented at the top.
- **Zwitterionic Dipolar Character:** In aqueous solution, internal acid-base neutralization occurs: the acidic carboxyl group ($-\mathrm{COOH}$) releases a proton, while the basic amino group ($-\mathrm{NH_2}$) accepts it. This forms a dipolar ion termed a **zwitterion**:

$$\mathrm{H_2N\text{--}CH(R)\text{--}COOH \rightleftharpoons H_3N^{+}\text{--}CH(R)\text{--}COO^{-}}$$

  Although electrically neutral, zwitterions possess distinct positive and negative charges. Consequently, amino acids behave as high-melting, water-soluble, crystalline salt-like solids with high dipole moments, unlike ordinary neutral amines or haloacids.
- **Amphoteric Behavior:** In acidic media ($\mathrm{pH < pI}$), the carboxylate ion accepts a proton, yielding a cation that migrates toward the cathode during electrophoresis. In basic media ($\mathrm{pH > pI}$), the ammonium group surrenders a proton, yielding an anion that migrates toward the anode:

$$\underset{\text{Cationic (in strong acid)}}{\mathrm{H_3N^{+}\text{--}CH(R)\text{--}COOH}} \xleftarrow[\mathrm{H^{+}}]{\mathrm{OH^{-}}} \underset{\text{Dipolar Zwitterion (neutral)}}{\mathrm{H_3N^{+}\text{--}CH(R)\text{--}COO^{-}}} \xrightarrow[\mathrm{H^{+}}]{\mathrm{OH^{-}}} \underset{\text{Anionic (in strong base)}}{\mathrm{H_2N\text{--}CH(R)\text{--}COO^{-}}}$$



Dipolar Zwitterionic Transformations and Amide Peptide Bond Formation via Dehydration Condensation

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3.2 Classification of Amino Acids

- **Acidic Amino Acids:** Contain more carboxyl groups than amino groups (e.g., Aspartic acid, Glutamic acid).
- **Basic Amino Acids:** Contain more basic nitrogenous/amino centers than carboxyl groups (e.g., Lysine, Arginine, Histidine).
- **Neutral Amino Acids:** Contain equal numbers of amino and carboxyl functions (e.g., Glycine, Alanine, Valine, Leucine).
- **Essential vs. Non-essential:**
  - **Non-essential amino acids:** Ten amino acids that can be biosynthesized de novo by human tissues; dietary supplementation is not mandatory.
  - **Essential amino acids:** Ten amino acids that human metabolic pathways cannot synthesize; they must be ingested regularly through food. These are: **Valine, Leucine, Isoleucine, Phenylalanine, Methionine, Tryptophan, Threonine, Lysine, Arginine, and Histidine**.

3.3 The Peptide Linkage

Proteins are built when α-amino acids link covalently through an amide bridge called a **peptide bond** ($-\mathrm{CO}\text{--}\mathrm{NH}\text{--}$). When the α-COOH of one amino acid condenses with the α-$\mathrm{NH_2}$ of another, a molecule of water is eliminated. Two joined amino acids constitute a *dipeptide* (e.g., Glycylalanine, Gly-Ala). Linking three yields a *tripeptide* (bearing two peptide bonds). Polypeptides possess more than ten amino acid residues. By convention, a polypeptide with over 100 residues and a molecular mass exceeding $10,000\,\mathrm{u}$ is categorized as a **protein** (though small proteins with well-defined 3D shapes, such as insulin with 51 amino acids, are also designated proteins).

3.4 Fibrous vs. Globular Proteins

Classification of Proteins Based on Molecular Shape

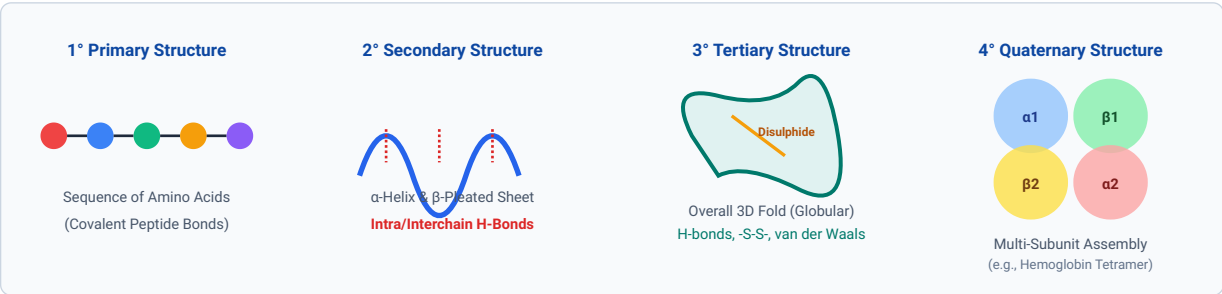
Characteristic	Fibrous Proteins	Globular Proteins
Polypeptide Chain Geometry	Linear chains running parallel to one another, forming tough elongated fibres.	Polypeptide chains fold and coil tightly back upon themselves into compact, spheroidal shapes.
Stabilizing Forces	Extensive intermolecular hydrogen bonds and covalent disulphide ($-\mathrm{S}\text{--}\mathrm{S}\text{--}$) cross-links.	Intramolecular hydrogen bonds, disulphide linkages, ionic attractions, and hydrophobic interactions.
Water Solubility	Insoluble in water.	Generally soluble in water or dilute saline solutions.
Biological Functions	Structural integrity, protection, mechanical tension.	Dynamic functions: metabolic catalysis, hormonal signaling, transport, antibodies.
Classic Examples	**Keratin** (hair, wool, nails, silk), **Myosin** (muscle fibers), Collagen.	**Insulin**, **Albumin** (egg white and serum), Hemoglobin.

3.5 Four Hierarchical Levels of Protein Structure

1. **Primary Structure (1°):** The precise linear sequence in which specific amino acids are covalently linked by peptide bonds. Every protein has a genetically specified, unique primary sequence. Altering even a single amino acid disrupts biological activity (e.g., substitution of glutamic acid by valine at position 6 of the β-globin chain causes sickle-cell anaemia).
2. **Secondary Structure (2°):** Local, regular conformational folding of the polypeptide backbone stabilized exclusively by hydrogen bonding between the carbonyl oxygen ($>\mathrm{C}=\mathrm{O}$) of one peptide group and the amide hydrogen ($-\mathrm{NH}\text{--}$) of another:
   - **α-Helix:** The polypeptide chain twists into a right-handed screw. Every $-\mathrm{NH}$ group forms an intramolecular hydrogen bond with the $>\mathrm{C}=\mathrm{O}$ situated four residues ahead in the adjacent turn of the helix. Side-chain R groups project outward.
   - **β-Pleated Sheet:** Polypeptide chains are stretched out to near-maximum extension and aligned side-by-side. Neighboring chains are cross-linked by intermolecular hydrogen bonds, creating a rippled drapery-like sheet.
3. **Tertiary Structure (3°):** The overall, three-dimensional spatial folding of secondary structural elements into compact globular or fibrous shapes. This structure is stabilized by four distinct non-covalent and covalent interactions:
   - Hydrogen bonds
   - Disulphide linkages ($-\mathrm{S}\text{--}\mathrm{S}\text{--}$ covalent bonds between cysteine residues)
   - Electrostatic ionic attractions (salt bridges between basic and acidic side chains)
   - Van der Waals forces and hydrophobic interactions
4. **Quaternary Structure (4°):** The spatial arrangement and assembly of two or more independent polypeptide subunits (individual tertiary folded chains) into a single functional, oligomeric protein complex. For example,

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haemoglobin is a tetramer composed of two identical α-subunits and two identical β-subunits arranged cooperatively around four iron-protoporphyrin haem groups.



Structural Hierarchy of Proteins: Primary Sequence, Secondary Folding, Tertiary 3D Architecture, and Quaternary Multimeric Assembly

3.6 Denaturation of Proteins

A protein existing in its natural biological environment with a specific three-dimensional conformation and active biological function is termed a **native protein**. When this native protein is exposed to physical changes (e.g., thermal heating) or chemical disturbances (e.g., pH shifts, mineral acids, organic solvents):

- Weak non-covalent hydrogen bonds, ionic salt bridges, and hydrophobic attractions are disrupted.
- Globular domains unfold, helical coils uncoil, and the precise tertiary and secondary conformations collapse.
- The protein precipitates or coagulates irreversibly, losing its biological activity completely.
- **Crucial NEET Point:** During denaturation, **secondary (2°) and tertiary (3°) structures are destroyed, but the primary (1°) structure remains completely intact** because strong covalent peptide bonds are not hydrolysed under moderate denaturing conditions.

Classic Manifestations of Denaturation:

1. **Coagulation of egg white on boiling:** The soluble globular protein albumin undergoes heat-induced denaturation, forming an insoluble, cross-linked fibrous matrix that entraps surrounding water.
2. **Curdling of milk:** Metabolic action of *Lactobacillus* bacteria produces lactic acid, dropping the pH. The acidic environment disrupts ionic stabilization in milk casein, causing it to coagulate.

4. Enzymes and Biocatalysis

Nearly all biological processes are coordinated by biochemical catalysts termed **enzymes**. Chemically, almost all enzymes are **globular proteins**. Enzymes exhibit exceptional catalytic power and absolute substrate- and reaction-specificity, operating under mild physiological temperatures ($\sim 310\,\mathrm{K}$) and neutral pH (~ 7.4).

4.1 Nomenclature of Enzymes

- Named after the specific substrate acted upon, replacing the suffix with **-ase**. For example, *maltase* catalyses the hydrolytic cleavage of maltose into two molecules of glucose:

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \xrightarrow{Maltase} 2C_6H_{12}O_6\ (Glucose)}$$

- Named according to the nature of the reaction catalysed. For instance, enzymes that catalyse simultaneous oxidation of one substrate and reduction of another are termed **oxidoreductases**.

4.2 Mechanism: Lowering of Activation Energy

Like inorganic catalysts, enzymes accelerate reactions by providing an alternate reaction pathway characterized by a markedly lower **activation energy (E_a)**. The substrate binds to a specific *active site* on the enzyme surface to form an enzyme-substrate complex, which undergoes reaction and releases products, regenerating the enzyme:

Quantitative Comparison: Hydrolysis of Sucrose

- Activation energy for ordinary mineral acid-catalysed hydrolysis: $E_a = 6.22\,\mathrm{kJ\,mol^{-1}}$.
- Activation energy for biological *sucrase*-catalysed hydrolysis: $E_a = 2.15\,\mathrm{kJ\,mol^{-1}}$.

By depressing the kinetic activation barrier by over 65%, sucrase accelerates sucrose cleavage by millions of times under physiological conditions.

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5. Vitamins

Vitamins are essential organic compounds required in small quantities in the diet to carry out vital physiological functions for the maintenance, normal growth, and health of the organism. Most vitamins cannot be synthesized endogenously by human cells (though intestinal gut microflora can generate limited quantities of certain B-complex vitamins and vitamin K). Deficiency of specific vitamins leads to well-characterized pathological deficiency syndromes.

5.1 Classification by Solubility

1. **Fat-Soluble Vitamins:**
   - Vitamins **A, D, E, and K**.
   - Soluble in fats, oils, and non-polar organic solvents, but insoluble in water.
   - Stored endogenously in the **liver** and **adipose tissues**. Hypervitaminosis can occur if taken in excessive amounts.

2. **Water-Soluble Vitamins:**
   - **B-group vitamins** ($\mathrm{B}_1, \mathrm{B}_2, \mathrm{B}_6, \mathrm{B}_{12}$, etc.) and **Vitamin C** (ascorbic acid).
   - Readily soluble in water.
   - Because they are readily excreted in urine, they **cannot be stored in the body** and must be supplied continuously via daily diet.
   - **High-Yield NEET Exception:** **Vitamin B_{12} (cyanocobalamin) is water-soluble, yet it is stored in significant reserves in the liver for several years.**

Comprehensive Master Chart of Vitamins, Dietary Sources, and Deficiency Syndromes

Vitamin & Chemical Name	Solubility	Primary Dietary Sources	Clinical Deficiency Diseases & Pathology
Vitamin A (Retinol)	Fat	Fish liver oil, carrots, butter, milk, yellow fruits.	**Xerophthalmia** (keratinization and hardening of the cornea); **Night blindness** (nyctalopia).
Vitamin B_1 (Thiamine)	Water	Yeast, whole milk, unpolished cereals, green vegetables.	**Beri-beri** (neurological weakness, extreme muscular wasting, loss of appetite, retarded growth).
Vitamin B_2 (Riboflavin)	Water	Milk, egg white, liver, kidney, leafy greens.	**Cheilosis** (painful fissures and cracks at the angles of mouth and lips); digestive disorders; burning skin sensations.
Vitamin B_6 (Pyridoxine)	Water	Yeast, milk, egg yolk, cereals, grams, legumes.	**Convulsions**, peripheral neuropathy, impaired amino acid metabolism.
Vitamin B_{12} (Cobalamin)	Water (Stored in liver)	Meat, fish, egg, curd (synthesized exclusively by bacteria; absent in unfortified plant foods).	**Pernicious anaemia** (macrocytic megaloblastic anaemia where red blood cells are deficient in haemoglobin; neurological demyelination).
Vitamin C (Ascorbic acid)	Water	Citrus fruits (lemon, orange), Indian gooseberry (amla), green leafy vegetables.	**Scurvy** (fragile capillaries, bleeding spongy gums, loose teeth, petechial haemorrhages, delayed wound healing).
Vitamin D (Calciferols)	Fat	Cutaneous photochemical synthesis under sunlight; fish liver oil, egg yolk, enriched milk.	**Rickets** in children (defective bone mineralization, bow legs, pigeon chest); **Osteomalacia** in adults (soft bones, skeletal pain, fractures).
Vitamin E (Tocopherols)	Fat	Vegetable oils (wheat germ oil, sunflower oil, cotton seed oil), nuts.	**Increased fragility of erythrocytes** (haemolytic anaemia); muscular dystrophy and neuromuscular weakness.
Vitamin K (Phylloquinones)	Fat	Green leafy vegetables (spinach, cabbage), synthesized by intestinal microflora.	**Increased blood clotting time** (delayed prothrombin synthesis leading to haemorrhagic tendencies).

6. Nucleic Acids and Molecular Genetics

The transmission of inherited characteristics from parent to progeny across generations is termed **heredity**. The nuclear structures orchestrating heredity are chromosomes, which are composed of proteins and high-molecular-weight polynucleotides called **nucleic acids**: **Deoxyribonucleic acid (DNA)** and **Ribonucleic acid (RNA)**.

6.1 Chemical Composition

Complete hydrolytic decomposition of any nucleic acid yields three chemical fragments:

1. **A Five-Carbon Pentose Sugar:**
   - In **RNA**, the sugar is β-D-ribose (possessing an $-\mathrm{OH}$ group at $\mathrm{C2'}$).
   - In **DNA**, the sugar is β-D-2-deoxyribose (lacking an oxygen atom at $\mathrm{C2'}$, possessing two $-\mathrm{H}$ atoms).

2. **Phosphoric Acid (H_3PO_4):** Forms ester linkages with sugar hydroxyl groups, creating the negatively charged polyanionic backbone.
3. **Nitrogenous Heterocyclic Bases:**
   - **Purines (Bicyclic): Adenine (A)** and **Guanine (G)** (present in both DNA and RNA).
   - **Pyrimidines (Monocyclic): Cytosine (C)** and **Thymine (T)** occur in DNA. In RNA, **Uracil (U)** replaces Thymine. Thus:

$$\text{DNA Bases: A, G, C, T} \qquad \text{RNA Bases: A, G, C, U}$$

6.2 Nucleosides vs. Nucleotides

- **Nucleoside:** A unit formed by the covalent attachment of a purine or pyrimidine base to the $C1'$ position of the pentose sugar via an N-glycosidic bond:

$$\text{Nucleoside} = \text{Pentose Sugar} + \text{Nitrogenous Base at } C1'$$

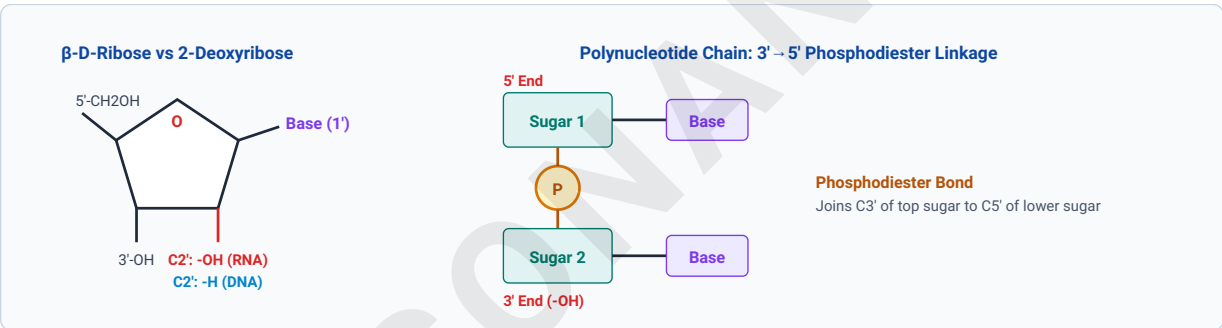
  (To avoid ambiguity, sugar carbons are numbered with primes: $1', 2', 3', 4', 5'$).
- **Nucleotide:** Formed when a nucleoside undergoes esterification with phosphoric acid at the $C5'$ hydroxyl group of the pentose sugar:

$$\text{Nucleotide} = \text{Nucleoside} + \text{Phosphate at } C5' = \text{Base} + \text{Sugar} + \text{Phosphate}$$

  Nucleotides serve as the monomeric repeat units of all polynucleotides.

6.3 The Phosphodiester Linkage

Nucleotides polymerize when the $5'$-phosphate group of one nucleotide condenses with the $3'$-hydroxyl group of the pentose sugar of an adjacent nucleotide. This bridging unit is termed a $3' \rightarrow 5'$ **phosphodiester linkage**. A polynucleotide chain has a definite directional polarity: a free $5'$-phosphate terminus at one end and an unesterified $3'$-hydroxyl terminus at the opposite end.



Structural Architecture of Nucleic Acids: Pentose Differentiation and $3' \rightarrow 5'$ Phosphodiester Backbone Continuity

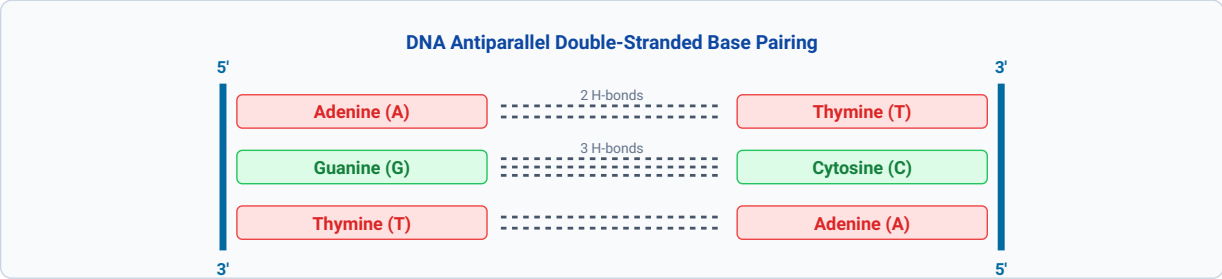
6.4 Watson-Crick Double-Helical Model of DNA

In 1953, James Watson and Francis Crick postulated the three-dimensional double-helix model of DNA:

- DNA consists of two polynucleotide strands coiling helically around a central common axis in a right-handed direction, resembling a twisted ladder.
- The two strands run in strictly **antiparallel** orientations: one strand is directed $5' \rightarrow 3'$, while the complementary strand is oriented $3' \rightarrow 5'$.
- The hydrophilic rails (backbone) of the ladder consist of alternating units of deoxyribose sugar and phosphate groups; the hydrophobic rungs are composed of paired planar nitrogenous bases pointing inward.
- **Complementary Base Pairing:** The bases from opposite strands pair through specific hydrogen bonding:
  - **Adenine pairs exclusively with Thymine** ($A = T$) via **two hydrogen bonds**.
  - **Guanine pairs exclusively with Cytosine** ($G \equiv C$) via **three hydrogen bonds**.

  Because of this stereochemical requirement, the base sequence of one strand automatically dictates the sequence of its partner.

BOSONANCE



Antiparallel Directionality and Watson-Crick Complementary Hydrogen-Bonding Topology: $\mathrm{A} = \mathrm{T}$ (2 H-Bonds) and $\mathrm{G} \equiv \mathrm{C}$ (3 H-Bonds)

6.5 Ribonucleic Acid (RNA) and Biological Roles

Unlike DNA, the secondary structure of RNA is predominantly **single-stranded**, though it can fold back upon itself into local hairpin loops. RNA molecules occur as three distinct functional types:

- **Messenger RNA (m-RNA):** Carries transcribed genetic instructions from nuclear DNA to cytoplasmic ribosomes.
- **Ribosomal RNA (r-RNA):** Structural and catalytic component of ribosomes, the site of protein translation.
- **Transfer RNA (t-RNA):** Adapter molecules that bind specific amino acids and align them with matching codons on the m-RNA template.

6.6 DNA Fingerprinting

Conventional dermal fingerprints can be cosmetically or surgically altered. However, the sequence of nucleotide base pairs in an individual's DNA is strictly unique (except in identical twins), identical in every somatic cell, and completely refractory to alteration by any known clinical or surgical intervention. Determining this sequence pattern is known as **DNA fingerprinting**. Major applications include:

- Forensic crime scene investigation to identify suspects.
- Paternity disputes and family lineage verification.
- Identification of severely disfigured victims in mass casualty incidents.
- Tracing racial origin, evolutionary migrations, and phylogenetic lineage.

7. Hormones

Hormones are biochemical molecules produced by ductless **endocrine glands** that function as **intercellular chemical messengers**. They are secreted directly into the bloodstream, which transports them to distant target cells and organs to maintain biological homeostasis, regulate metabolism, mediate stress responses, and direct growth and sexual development.

7.1 Chemical Classification

Classification of Endocrine Hormones Based on Chemical Nature

Chemical Category	Representative Hormones	Endocrine Gland & Key Physiological Function
Steroid Hormones (Lipophilic cyclopentanoperhydrophenanthrene ring derivatives)	**Estrogens, Estradiol** **Progesterone** **Testosterone (Androgens)** **Glucocorticoids (Cortisone, Cortisol)** **Mineralocorticoids (Aldosterone)**	**Ovaries:** Female secondary sex characters; menstrual cycle. **Corpus Luteum:** Prepares uterine endometrium for blastocyst implantation. **Testes:** Male secondary sex characters (deep voice, facial hair). **Adrenal Cortex:** Carbohydrate metabolism, stress response, anti-inflammatory modulation. **Adrenal Cortex:** Controls renal excretion of water and salt ($\mathrm{Na}^{+}/\mathrm{K}^{+}$).
Polypeptide / Protein Hormones (Hydrophilic polymers of amino acids)	**Insulin** (51 amino acids) **Glucagon** **Endorphins**	**Pancreas (β-cells):** Lowers blood glucose by facilitating cellular uptake and glycogen synthesis. **Pancreas (α-cells):** Raises blood glucose by stimulating glycogenolysis. **Pituitary / Brain:** Endogenous analgesia, pain modulation.
Amino Acid Derivatives (Chemically modified single amino acids)	**Epinephrine (Adrenaline)** **Norepinephrine** **Thyroxine (T_4)**	**Adrenal Medulla:** "Fight or flight" response to immediate external stress. **Adrenal Medulla:** Vasopressor and neurotransmitter functions.

BOSONANCE

Chemical Category	Representative Hormones	Endocrine Gland & Key Physiological Function
		Thyroid Gland: Iodinated derivative of the amino acid **tyrosine**; regulates basal metabolic rate.

7.2 Clinical Pathophysiology of Endocrine Glands

- **Blood Glucose Regulation: Insulin** and **glucagon** act antagonistically. A rapid spike in blood glucose prompts insulin release, driving glucose conversion to glycogen. When blood sugar drops, glucagon stimulates glycogen breakdown to maintain blood glucose within narrow physiological limits ($70\text{--}100\,\mathrm{mg/dL}$ fasting).
- **Thyroid Pathophysiology: Thyroxine** is an iodinated derivative of tyrosine.
  - **Hypothyroidism:** Abnormally low thyroxine production, characterized by lethargy, low basal metabolic rate, and obesity. Dietary iodine deficiency causes compensatory thyroid enlargement, known as **goitre**. This condition is controlled by adding sodium iodide to commercial table salt ("iodised salt").
  - **Hyperthyroidism:** Excessive secretion of thyroxine, resulting in elevated metabolic rate, weight loss, and nervous excitability.
- **Adrenal Cortex Pathophysiology:** The adrenal cortex releases glucocorticoids (regulating carbohydrate, lipid, and protein metabolism and modulating inflammation) and mineralocorticoids (regulating renal salt and water reabsorption). Dysfunction or destruction of the adrenal cortex leads to **Addison's disease**, characterized by severe hypoglycemia, chronic asthenia, hyperpigmentation, and heightened susceptibility to stress. Addison's disease is fatal unless treated with replacement mineralocorticoid and glucocorticoid therapy.

8. Worked Examples: Step-by-Step NEET Problem Solving

Worked Example 1: Deducing Monosaccharide Reaction Stoichiometry

Problem: An unknown open-chain aldohexose X has the molecular formula $\mathrm{C_6H_{12}O_6}$. Upon prolonged heating with concentrated HI, it produces compound Y. When treated with excess acetic anhydride in pyridine, $1\,\mathrm{mol}$ of X reacts with $5\,\mathrm{mol}$ of acetic anhydride to yield derivative Z. Mild oxidation of X with bromine water yields monocarboxylic acid W, which upon oxidation with hot concentrated $\mathrm{HNO_3}$ produces dicarboxylic acid U. Identify compounds Y, Z, W, and U, and explain why derivative Z fails to form an oxime upon treatment with $\mathrm{NH_2OH}$.

Step-by-Step Solution:

1. Heating an aldohexose with HI causes complete reductive deoxygenation of the 6-carbon skeleton to produce unbranched n-**hexane** ($\mathrm{CH_3\text{--}(CH_2)_4\text{--}CH_3}$). Therefore, compound Y is n-hexane.
2. Reaction of $1\,\mathrm{mol}$ of hexose with $5\,\mathrm{mol}$ of acetic anhydride demonstrates the presence of five hydroxyl groups on separate carbon atoms. Derivative Z is **glucose pentaacetate**.
3. Bromine water is a selective, mild oxidising agent that oxidises only terminal aldehyde groups without attacking primary or secondary alcohols. Hence, compound W is **gluconic acid** ($\mathrm{HOOC\text{--}(CHOH)_4\text{--}CH_2OH}$).
4. Concentrated nitric acid oxidises both the terminal aldehyde and the primary alcohol ($-\mathrm{CH_2OH}$) to carboxyl groups. Hence, compound U is the dicarboxylic acid **saccharic acid** ($\mathrm{HOOC\text{--}(CHOH)_4\text{--}COOH}$).
5. **Absence of Oxime Formation:** In glucose pentaacetate (Z), the anomeric oxygen at $\mathrm{C1}$ is esterified into a stable cyclic hemiacetal acetate. Because the anomeric $\mathrm{C1}$ is locked, it cannot open into a free, unhindered aldehyde group. Consequently, it cannot react with hydroxylamine to form an oxime.

Worked Example 2: Inversion of Cane Sugar and Optical Rotation

Problem: Native sucrose is dextrorotatory with a specific optical rotation of $+66.5^\circ$. When subjected to hydrolytic cleavage catalysed by dilute acid or the enzyme invertase, the resulting aqueous solution turns laevorotatory (-19.95°). Explain this optical inversion chemically using specific optical rotation values for the hydrolysis products.

Step-by-Step Solution:

Sucrose is a non-reducing disaccharide composed of α-D-glucopyranose and β-D-fructofuranose joined by an $\alpha\text{-}1 \leftrightarrow \beta\text{-}2$ glycosidic linkage.

Hydrolysis cleaves this glycosidic linkage, releasing an equimolar ($1:1$) mixture of D-(+)-glucose and D-(-)-fructose:

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6\ (\text{D-(+)-glucose}) + C_6H_{12}O_6\ (\text{D-(-)-fructose})}$$

- Specific optical rotation of D-(+)-glucose: $[\alpha]_{\mathrm{D}} = +52.5^\circ$ (dextrorotatory).
- Specific optical rotation of D-(-)-fructose: $[\alpha]_{\mathrm{D}} = -92.4^\circ$ (strongly laevorotatory).

The net observed optical rotation of the equimolar product mixture is the average of these two values:

$$[\alpha]_{\text{mixture}} = \frac{+52.5^\circ + (-92.4^\circ)}{2} = \frac{-39.9^\circ}{2} = -19.95^\circ$$

Because the magnitude of laevorotation of D-(-)-fructose (-92.4°) significantly exceeds the dextrorotation of D-(+)-glucose ($+52.5^\circ$), the sign of optical rotation inverts from dextro ($+$) to laevo ($-$). The product is therefore called **invert sugar**.

Worked Example 3: Polynucleotide Base Pairing and Chargaff's Equivalence

Problem: Quantitative analysis of a sample of double-stranded DNA demonstrates that 32% of the total nitrogenous bases are Cytosine. Determine the percentage composition of Adenine, Thymine, and Guanine in this DNA sample. Would this quantitative relationship hold if the sample were hydrolysed single-stranded RNA?

Step-by-Step Solution:

1. In double-stranded DNA, Watson-Crick base pairing requires that Guanine pairs exclusively with Cytosine via three hydrogen bonds ($\mathrm{G} \equiv \mathrm{C}$). Therefore:
$$\%\,\text{Guanine} = \%\,\text{Cytosine} = 32\%$$
2. The combined percentage of $\mathrm{G} + \mathrm{C}$ is:
$$32\% + 32\% = 64\%$$
3. The remaining percentage belongs to the $\mathrm{A} = \mathrm{T}$ base pairs:
$$\%\,(\text{Adenine} + \text{Thymine}) = 100\% - 64\% = 36\%$$
4. Because Adenine pairs exclusively with Thymine ($\mathrm{A} = \mathrm{T}$), their molar concentrations are identical:
$$\%\,\text{Adenine} = \%\,\text{Thymine} = \frac{36\%}{2} = 18\%$$
5. **RNA Consideration:** If the sample were single-stranded RNA, **no fixed molar equivalence exists** between bases ($\mathrm{A} \neq \mathrm{U}$ and $\mathrm{G} \neq \mathrm{C}$). In single-stranded RNA, nucleotides are not paired along an antiparallel complementary strand. Therefore, the base ratios vary unpredictably.

9. NEET-UG Practice Assessment Section

The following original practice questions match the format, cognitive level, and difficulty of the NEET-UG examination (single-correct MCQs, assertion-reason, two-statement evaluations, and matching tables).

BOSONANCE

Practice Question 1

Single Correct MCQ

Which of the following chemical reactions definitively confirms that the six carbon atoms in D-glucose form an unbranched, straight chain?

A. Oxidation with bromine water to form gluconic acid

B. Formation of an oxime with hydroxylamine

C. Prolonged heating with excess hydrogen iodide (HI) and red phosphorus to yield n-hexane

D. Reaction with acetic anhydride to produce a pentaacetate derivative

Correct Answer: C

Worked Scientific Rationale: Prolonged heating of glucose with HI causes complete reductive removal of all oxygen functions, yielding n-hexane. Because n-hexane is an unbranched hydrocarbon, this reaction proves all six carbon atoms are joined in a continuous, straight chain. Option A confirms that the carbonyl is an aldehyde; Option B confirms the presence of a carbonyl group; Option D demonstrates the existence of five hydroxyl groups on separate carbon atoms.

Practice Question 2

Single Correct MCQ

Which of the following statements regarding the structural features of sucrose is correct?

A. It is composed of β-D-glucopyranose and α-D-fructofuranose linked by a β-1,4-glycosidic bond.

B. It reduces Fehling's solution and exhibits mutarotation in aqueous solution.

C. The glycosidic bond links $\mathrm{C1}$ of α-D-glucopyranose to $\mathrm{C2}$ of β-D-fructofuranose, making it a non-reducing sugar.

D. Hydrolysis of dextrorotatory sucrose yields an equimolar product mixture that remains dextrorotatory.

Correct Answer: C

Worked Scientific Rationale: Sucrose is composed of α-D-glucopyranose and β-D-fructofuranose. The glycosidic linkage forms between the anomeric $\mathrm{C1}$ of glucose and the anomeric $\mathrm{C2}$ of fructose. Because both reducing groups are locked in this bond, sucrose cannot generate a free carbonyl group in solution, making it non-reducing and incapable of mutarotation. Hydrolysis yields an equimolar mixture of glucose ($[\alpha]_{\mathrm{D}} = +52.5^{\circ}$) and fructose ($[\alpha]_{\mathrm{D}} = -92.4^{\circ}$), producing a net laevorotatory solution (-19.95°).

BOSONANCE

Practice Question 3

Single Correct MCQ

During the thermal or chemical denaturation of a globular protein, which structural level remains intact?

A. Primary structure

B. Secondary structure

C. Tertiary structure

D. Quaternary structure

Correct Answer: A

Worked Scientific Rationale: Denaturation disrupts non-covalent hydrogen bonds, ionic interactions, and hydrophobic contacts, causing unfolding of the tertiary globule and uncoiling of secondary helices and pleated sheets. However, the covalent peptide bonds connecting the amino acids in the primary sequence remain unbroken. Thus, the primary structure remains completely intact.

Practice Question 4

Assertion & Reason

Assertion (A): Unlike haloacids or simple amines, α-amino acids exist as high-melting, crystalline, water-soluble solids with salt-like character.
Reason (R): In aqueous solution, the carboxyl group loses a proton and the amino group accepts a proton, forming an internal dipolar zwitterion.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Correct Answer: A

Worked Scientific Rationale: Amino acids contain both an acidic carboxyl group ($-\mathrm{COOH}$) and a basic amino group ($-\mathrm{NH_2}$). Internal proton transfer forms a dipolar zwitterion ($\mathrm{H_3N^+}\text{--}\mathrm{CH(R)}\text{--}\mathrm{COO^-}$). Strong electrostatic attractions between these opposite charges produce a stable crystal lattice, resulting in high melting points and high water solubility. Both statements are true, and (R) correctly explains (A).

BOSONANCE

Practice Question 5

Statement Evaluation

Statement I: Water-soluble vitamins must be supplied regularly in the daily human diet because they are readily excreted in urine and cannot be stored in the body.
Statement II: Vitamin B_{12} is an exception among water-soluble vitamins because it is stored in the human liver for several years.

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Correct Answer: A

Worked Scientific Rationale: Water-soluble vitamins (B-complex and C) dissolve in water and are continuously filtered and excreted by the kidneys. Consequently, they cannot be stored in large quantities and must be supplied regularly in the diet (Statement I is correct). Vitamin B_{12} (cobalamin) is a well-established biological exception; it binds to intrinsic factor and is stored in significant amounts in the liver (Statement II is correct).

Practice Question 6

Match the Column

Match the vitamins listed in Column I with their clinical deficiency diseases in Column II:

Column I (Vitamin)	Column II (Deficiency Pathology)
(a) Vitamin B_1	(i) Cheilosis and fissuring of mouth corners
(b) Vitamin B_2	(ii) Beri-beri
(c) Vitamin B_{12}	(iii) Increased blood clotting time
(d) Vitamin K	(iv) Pernicious anaemia

A. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

B. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

C. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

D. (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

Correct Answer: A

Worked Scientific Rationale: Vitamin B_1 (thiamine) deficiency causes Beri-beri (a)-(ii). Vitamin B_2 (riboflavin) deficiency causes Cheilosis (b)-(i). Vitamin B_{12} deficiency causes Pernicious anaemia (c)-(iv). Vitamin K is required for prothrombin biosynthesis; its deficiency increases blood clotting time (d)-(iii).

BOSONANCE

Practice Question 7

Single Correct MCQ

Which of the following amino acids is achiral (optically inactive)?

A. Alanine

B. Glycine

C. Valine

D. Leucine

Correct Answer: B

Worked Scientific Rationale: The general formula of an α-amino acid is $\mathrm{R\text{--}CH(NH_2)\text{--}COOH}$. In glycine, the side chain is hydrogen ($\mathrm{R = H}$). The α-carbon is bonded to two identical hydrogen atoms ($\mathrm{H_2N\text{--}CH_2\text{--}COOH}$), making it symmetric (achiral). All other 19 natural amino acids contain four distinct groups on the α-carbon and are chiral.

Practice Question 8

Single Correct MCQ

The glycosidic linkage connecting the monosaccharide units in amylose is:

A. β-1,4-glycosidic linkage

B. α-1,4-glycosidic linkage

C. α-1,6-glycosidic linkage

D. β-1,6-glycosidic linkage

Correct Answer: B

Worked Scientific Rationale: Amylose is the water-soluble component of starch ($15\text{--}20\%$). It consists of a long, unbranched chain of $200\text{--}1000$ α-D-(+)-glucose units linked exclusively by α-1,4-glycosidic linkages. Branch points in amylopectin and glycogen involve α-1,6-linkages, while cellulose contains β-1,4-glycosidic bonds.

BOSONANCE

Practice Question 9

Single Correct MCQ

Which heterocyclic nitrogenous base is present in DNA but absent in RNA?

A. Uracil

B. Cytosine

C. Thymine

D. Guanine

Correct Answer: C

Worked Scientific Rationale: DNA contains Adenine (A), Guanine (G), Cytosine (C), and Thymine (T). RNA contains Adenine (A), Guanine (G), Cytosine (C), and Uracil (U). Thymine (5-methyluracil) is unique to DNA, whereas Uracil is found in RNA.

Practice Question 10

Assertion & Reason

Assertion (A): Glucose pentaacetate does not react with hydroxylamine to form an oxime.
Reason (R): The formation of glucose pentaacetate locks the cyclic hemiacetal pyranose ring, preventing mutarotation and the generation of an open-chain aldehyde group.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Correct Answer: A

Worked Scientific Rationale: In free D-glucose, the cyclic hemiacetal exists in dynamic equilibrium with a trace amount of the open-chain form, which reacts with hydroxylamine. Acetylating all five hydroxyl groups converts the anomeric $-\mathrm{OH}$ at $\mathrm{C1}$ into an ester (acetate). This locks the ring and prevents opening into the free carbonyl form. As a result, glucose pentaacetate cannot form an oxime. Both statements are true and (R) correctly explains (A).

BOSONANCE

Practice Question 11

Single Correct MCQ

Which of the following pairs of hormones are steroid derivatives secreted by the adrenal cortex and gonads?

A. Insulin and Glucagon

B. Epinephrine and Thyroxine

C. Testosterone and Glucocorticoids

D. Endorphins and Norepinephrine

Correct Answer: C

Worked Scientific Rationale: Testosterone and estradiol are steroid hormones produced by the gonads, and glucocorticoids (e.g., cortisol) and mineralocorticoids are steroid hormones produced by the adrenal cortex. Insulin and glucagon are polypeptides; epinephrine and norepinephrine are amino acid derivatives; thyroxine is an iodinated tyrosine derivative.

Practice Question 12

Single Correct MCQ

A nucleoside differs from a nucleotide in that a nucleoside lacks which of the following chemical components?

A. Heterocyclic base

B. Aldopentose sugar

C. Phosphate group

D. Hydroxyl group at $\mathrm{C3'}$

Correct Answer: C

Worked Scientific Rationale: A nucleoside is formed by attaching a nitrogenous base to the $\mathrm{C1'}$ carbon of a pentose sugar via an N-glycosidic bond. A nucleotide is formed when this nucleoside is esterified with phosphoric acid at the $\mathrm{C5'}$ position. Therefore, a nucleoside lacks the phosphate moiety.

BOSONANCE

Practice Question 13

Statement Evaluation

Statement I: All natural amino acids obtained from protein hydrolysis, with the exception of glycine, exhibit D-stereochemical configuration.
Statement II: In the α-helical secondary structure of proteins, the polypeptide backbone is stabilized by intramolecular hydrogen bonds between the $-\mathrm{NH}$ group of each amino acid and the $>\mathrm{C}=\mathrm{O}$ of an adjacent turn.

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Statement I is correct but Statement II is incorrect.

Correct Answer: C

Worked Scientific Rationale: Statement I is incorrect because nearly all naturally occurring amino acids have the **L-configuration** (with the $-\mathrm{NH}_2$ group on the left in Fischer projections), not the D-configuration. Statement II is correct: in an α-helix, the right-handed coiled backbone is stabilized by intramolecular hydrogen bonds between the $-\mathrm{NH}$ of one residue and the $>\mathrm{C}=\mathrm{O}$ of a residue four positions earlier along the chain.

Practice Question 14

Single Correct MCQ

Which of the following clinical conditions results from adrenal cortex insufficiency and is characterized by hypoglycemia, asthenia, and increased susceptibility to stress?

A. Xerophthalmia

B. Addison's disease

C. Cheilosis

D. Pernicious anaemia

Correct Answer: B

Worked Scientific Rationale: Impaired secretion of glucocorticoids and mineralocorticoids by the adrenal cortex leads to **Addison's disease**, which presents with hypoglycemia, chronic weakness, and decreased stress tolerance. Xerophthalmia is caused by vitamin A deficiency; cheilosis by vitamin B_2 deficiency; and pernicious anaemia by vitamin B_{12} deficiency.

BOSONANCE

Practice Question 15 Single Correct MCQ

How many peptide bonds are present in an octapeptide synthesized from eight individual α-amino acid molecules?

A. 8

B. 7

C. 6

D. 9

Correct Answer: B

Worked Scientific Rationale: A peptide containing n amino acid residues formed by linear condensation contains $(n-1)$ peptide bonds. For an octapeptide ($n=8$), the number of peptide bonds is $8-1=7$.

10. Common Student Pitfalls & High-Yield Revision Checklist

Critical Conceptual Traps in NEET Biomolecules

- **Confusing D/L Configuration with Optical Rotation $(+/-)$:** The prefix D or L denotes spatial stereochemical configuration relative to D-(+)-glyceraldehyde at the highest-numbered chiral carbon. It does not dictate whether a substance rotates plane-polarized light clockwise $(+)$ or counterclockwise $(-)$. For example, natural fructose has the D-configuration but is strongly laevorotatory, written as D-(-)-fructose.
- **Open-Chain vs. Cyclic Reducing Activity:** Fructose is a ketohexose, yet it acts as a reducing sugar because its α-hydroxyketone group tautomerizes to an aldose via an enediol intermediate in the alkaline media used for Tollens' and Fehling's tests.
- **Reducing vs. Non-reducing Disaccharides:** Sucrose is non-reducing because both anomeric carbons ($\mathrm{C1}$ of glucose and $\mathrm{C2}$ of fructose) form the glycosidic bond. In contrast, maltose and lactose leave one anomeric carbon as a free hemiacetal, retaining their reducing capacity.
- **Denaturation Scope:** Denaturation breaks secondary and tertiary hydrogen bonds, hydrophobic interactions, and salt bridges, but leaves primary covalent peptide bonds intact.
- **Vitamin B_{12} Solubility vs. Storage:** Vitamin B_{12} is water-soluble, but unlike other water-soluble vitamins, it is stored in the liver for extended periods.

Carbohydrates Mastery:
Remember that glucose + Br_2 water yields gluconic acid (monocarboxylic), while glucose + concentrated HNO_3 yields saccharic acid (dicarboxylic).

Amino Acid Configurations:
All natural α-amino acids from proteins have the L-configuration. Glycine is the only achiral amino acid.

Enzyme Energetics:
Enzymes lower the activation energy barrier (E_a) without altering the overall thermodynamic equilibrium (ΔG°).

Nucleotide Linkages:
Sugars connect to bases at $\mathrm{C1}'$ via an N-glycosidic bond; adjacent nucleotides link from $\mathrm{C3}'$ to $\mathrm{C5}'$ through a phosphodiester bridge.

BOSONANCE

11. Quick High-Yield Recap

- **Glucose Reactions:** Reaction with HI yields n-hexane; $\mathrm{NH_2OH}$ yields an oxime; HCN yields a cyanohydrin; $\mathrm{Br_2/H_2O}$ yields gluconic acid; acetic anhydride yields a pentaacetate; concentrated $\mathrm{HNO_3}$ yields saccharic acid.
- **Cyclic Anomers:** In α-D-glucopyranose, the anomeric $\mathrm{C1\text{-}OH}$ points downward in Haworth projection; in β-D-glucopyranose, it points upward.
- **Disaccharide Bonds:** Sucrose has an $\alpha\text{-}1 \leftrightarrow \beta\text{-}2$ link (non-reducing); maltose has an $\alpha\text{-}1,4$ link (reducing); lactose has a $\beta\text{-}1,4$ link (reducing).
- **Polysaccharide Bonds:** Amylose has $\alpha\text{-}1,4$ linkages; amylopectin and glycogen have $\alpha\text{-}1,4$ linear links with $\alpha\text{-}1,6$ branches; cellulose has $\beta\text{-}1,4$ linkages.
- **Proteins:** Primary structure = amino acid sequence; secondary = α-helices and β-sheets (H-bonding); tertiary = overall 3D shape (fibrous/globular); quaternary = multi-subunit assembly.
- **Nucleic Acids:** Purines = Adenine, Guanine; Pyrimidines = Cytosine, Thymine (DNA), Uracil (RNA). In DNA, $\mathrm{A} = \mathrm{T}$ (2 H-bonds) and $\mathrm{G} \equiv \mathrm{C}$ (3 H-bonds).
- **Hormones:** Steroids (estrogens, androgens, glucocorticoids, mineralocorticoids); Polypeptides (insulin, glucagon); Amino acid derivatives (epinephrine, thyroxine).

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